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A dual-loop collaborative framework for smart construction sites

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Abstract
This study addresses the challenges of real-time response and closed-loop decision-making in smart construction. We propose a dual-loop collaborative framework integrating Building Information Modeling (BIM), the Internet of Things (IoT), and LSTM-Attention networks, validated via a 261-sensor deployment in the Taipei Bridge and Tunnel Joint Project. The architecture achieves sub-second synchronization and enables multi-objective optimization using NSGA-II. Experimental results demonstrate statistically significant improvements ($p < 0.001$): a 46.2% reduction in system latency, a 41.4% decrease in safety incidents, an 18.5% reduction in direct costs, and an 11.1% decline in CO_2 emissions, while maintaining a 99.1% safety compliance rate. The framework successfully bridges the gap between digital models and physical realization, providing a scalable solution for adaptive and sustainable construction management.

Keywords Smart construction, Digital twin, BIM-IoT, LSTM-attention, Multi-objective optimization

1 Introduction

Intelligent construction is undergoing a key transformation from "endpoint perception" to "overall collaboration" [1]. This transformation is critically important for the construction industry, which not only contributes about 40% of the global carbon emissions [2], but also faces severe challenges such as high energy consumption, low efficiency and frequent safety accidents [3, 4]. Although the deep integration of building information modeling (BIM), Internet of Things and artificial intelligence is regarded as the core path to improve the transparency of construction and the agility of decision-making [5], and promotes the evolution from "one-way monitoring" to "digital twin collaboration" [1], practical problems such as sensor noise, model heterogeneity and decision-making delay remain prominent.

Taking the Taipei Bridge and Tunnel Joint Project in 2024 as an example, although more than 260 sensors were deployed on site, the average delay of BIM updating and real-time decision-making exceeded 13 s [3]. Moreover, the manual recheck during the peak hours of crane operation will increase the delay by 5 to 7 s, which will seriously endanger the "golden window period" of safety intervention [6], highlighting the defects

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of one-way data flow in fast-paced construction [1, 7]. Existing research has several core limitations, including a lack of real-time capability, limited attention to end-to-end delay, packet loss compensation, and prediction algorithms [3, 8]. At the same time, there are issues such as insufficient closed-loop integrity, with control strategies mostly relying on manual scheduling or static rules, and a lack of adaptive "prediction-execution-feedback" mechanisms [5, 9]. In addition, cost, schedule and safety objectives are often optimized in isolation, lacking unified indicators and dynamic trade-off analysis, which makes it difficult to achieve multi-objective coordination [10, 11]. The disconnection between the digital model and the physical realization seriously restricts the sub-second response ability required by the high-density construction environment.

Based on the above findings, a new dual-loop collaborative framework is designed and verified in this study. The architecture realizes real-time synchronization through the data loop of sensor → gateway → MQTT → BIM, and realizes closed-loop control with the help of time series prediction [12]. This mechanism not only shortens the round-trip time of information between physical domain and digital domain, but also uses deep attention network to predict the construction progress, equipment health and environmental safety in real time, thus realizing multi-objective parallel optimization [13, 14]. It is worth noting that the dual-loop architecture we designed is scalable, allowing sensor nodes to be added and enabling additional business rules to be integrated into decision-making processes [15, 16].

This study aims to clarify the semantic interoperability mechanism of dual-loop architecture, and define the field mapping and timestamp synchronization strategy between edge gateway, MQTT proxy and IFC entity. Secondly, the collaborative advantages of the framework in delay convergence, risk identification and economic benefits are verified by experimental data. Finally, the quantitative relationship between interface delay, packet loss rate and security events can be summarized, which provides practical guidance for the implementation of large-scale intelligent construction sites [17, 18].

2 Literature review

2.1 Technological foundations and persistent gaps in enabling components

The intelligent transformation of the construction industry is driven by the integration of dual carbon targets and core digital technologies. The integration of BIM and Internet of Things (IoT) is the key to create digital twins, with research progressing from semantic mapping [19] to practical synchronization protocol [5] and spatial integration [3]. At the same time, deep learning models (such as LSTM network to enhance attention mechanism) have also made breakthroughs in equipment and process risk prediction [20, 21], and multi-objective optimization algorithms (such as NSGA-II and MOEA) have evolved to include sustainable development goals such as carbon emissions [22, 23].

However, there are still significant gaps in these components. BIM-IOT integration usually lacks real-time, and the reporting delay exceeds the critical threshold [3], preventing the full incorporation of environmental data. Predictive models usually analyze a single risk factor independently, ignoring the coupling between equipment operation and environmental conditions. Optimization algorithms mainly rely on static data, and rarely achieve integrated dynamic optimization of safety, cost, schedule and environmental impact within a unified framework.

2.2 Systemic architectural challenges: from cloud-edge deployment to closed-loop integration

In addition to individual technology, agile computing architecture is needed to realize real-time closed-loop system. Edge computing provides a low-latency basis for responsive decision-making [15, 16]. However, the current mainstream system architecture usually limits the function of edge devices to data preprocessing. This not only fails to fully tap the potential of localized intelligent processing but also fails to establish a robust and fault-tolerant cloud-edge collaborative mechanism to achieve uninterrupted control. This reveals a key systemic gap: although various component technologies (such as BIM, Internet of Things, and edge computing) are constantly improving, the overall architecture framework that can seamlessly integrate and operate together is still lacking. Specifically, the lack of real-time semantic synchronization, the lack of dynamic closed-loop control, and the limitations of extensible cloud-side cooperation mechanism all constitute the bottleneck of the core architecture layer. These shortcomings hinder the fundamental transformation of intelligent building systems from isolated technical applications to collaborative network physical systems that can coherently complete "perception-analysis-decision-execution".

2.3 Evolution of multi-objective optimization research: from economic scheduling to sustainability synergy

Facing the challenge of multi-objective decision-making in construction management, multi-objective optimization algorithm provides an effective solution tool. Many studies, such as Nusen et al. [17], combine genetic algorithm with BIM to schedule renovation projects, while Alothaimeen et al. use non-dominated sorting genetic algorithm II (NSGA-II) to optimize the cost and sustainability scores of green building projects [23]. Besides the classic NSGA-II, advanced optimization algorithms such as multi-objective ant colony optimizer (MOALO) and multi-objective equilibrium optimizer (MOEO) show excellent convergence and coverage in solving complex engineering problems. These provide additional algorithm options for dealing with nonlinear multi-modal optimization objectives in buildings [11, 24]. In addition, recent research attempts to combine more advanced algorithms such as NSGA-III with BIM to solve the optimization problem of sustainable design involving more objectives [25], showing the potential of algorithm evolution in solving complex collaborative challenges. Recently, the research scope has been expanded to include the environmental aspect. For example, Weigert et al. established a method to calculate carbon emissions during construction [2], while Zhao et al. simultaneously optimized the project duration, cost and carbon emissions in highway construction progress [22]. These studies have proved the potential of multi-objective optimization in promoting low-carbon construction and sustainable resource allocation [26–28]. This change marks a change from pure economic efficiency to broader value pursuit including environmental responsibility. However, a common limitation still exists: most optimization models rely on static or historical data and cannot be deeply integrated with real-time BIM Internet of Things data stream to respond to dynamic changes. In addition, safety objectives are often dealt with separately or simply combined, lacking systematic trade-offs within the same optimization framework as cost, schedule and carbon emissions. With regard to scheduling methods, besides genetic algorithm, new meta-heuristic algorithms, such as Sine-Cosine Algorithm

(SCA), are explored, and the construction sequence is generated by BIM, which shows the potential to meet the space-time constraints [29]. This research directly solves this challenge. We built a five-dimensional evaluation system at the decision-making rules level, including direct cost, safety indicators, project duration, carbon emissions and return on investment (ROI). The system adopts NSGA-II algorithm for real-time multi-objective optimization, and automatically identifies the Pareto optimal balance among economic efficiency, safety and environmental performance in dynamic construction environment. This represents a leap from single-objective or double-objective optimization to multi-dimensional collaborative decision-making.

2.4 Synthesis and positioning of the proposed framework

In short, the literature reveals an obvious disconnect: although enabling technologies have made progress, integrating them into a unified, adaptable, versatile, and real-time operating system remains an unsolved challenge. These limitations are not only technical but also architectural. Clearly, an overall system design that can seamlessly connect data, models, and decisions is needed. In this study, a dual-loop (data-control) collaborative architecture is proposed to solve this systematic gap identified above. The framework aims to fill the gaps in real-time synchronization, closed-loop optimization, semantic interoperability and elastic cloud-edge execution, and is positioned as a comprehensive solution for the next generation of intelligent construction sites. Accordingly, to further clarify the academic positioning of the proposed dual-loop collaborative framework, we systematically articulate its novelty by comparing it with existing BIM-IoT and digital twin architectures, with a particular focus on the fundamental innovations at the architectural level that distinguish it from traditional designs.

2.5 Novelty of the proposed dual-loop architecture

To clarify the framework's novelty and state-of-the-art position, this subsection compares it with typical BIM-IoT/digital twin architectures and elaborates on core innovations (Table 1). Most existing architectures use single-loop synchronization or separate modules, leading to real-time/adaptability trade-offs. In contrast, the framework has three key innovations: data-control loop decoupling/dynamic coupling (real-time synchronization + intelligent decision-making), elastic cloud-edge collaboration with fault-tolerant rollback (distributed computing), and real-time ifcXML/SensorML semantic alignment (resolving inconsistency), distinguishing it from traditional architectures and offering unique academic and technical value.

3 Materials and methods

3.1 Dual-loop collaborative framework

In the dynamic and high-noise environment of construction site, data undergoes multi-layer protocol conversion and semantic mapping in the round-trip process from sensors to decision nodes. Any delay may amplify the security risk. To overcome this bottleneck, we present the design of a parallel architecture that integrates a Data Loop and a Control Loop. The design is driven by the principle of separation of concerns, where the Data Loop ensures reliable state synchronization and the Control Loop handles intelligent analysis. As shown in Fig. 1, Data_Loop establishes millisecond-level flow channel through edge gateway and MQTT Broker. BIM server creates a one-to-one mapping

Table 1 Comparison of the proposed dual-loop framework with representative architectures

Architecture	Architecture type	Data flow	Decision-making mode	Latency control	Optimization objectives
Proposed dual-loop framework	Dual-loop (data-control) collaborative; cloud-edge fault-tolerant	Sensor → gateway → BIM + analysis → decision → executor; semantic alignment	Algorithm-driven automatic control; closed-loop feedback	Edge preprocessing + MQTT; sub-second response	Cost, safety, duration, carbon emissions, ROI (five-dimensional integration)
Skibniewski [1]	Construction 4.0/5.0; BIM + IoT+robotics + AI	Multi-source data → cloud → application	AI-assisted scheduling; human-robot collaboration	IoT real-time transmission; no sub-second design	Efficiency, safety, environmental sustainability, disaster resilience
Hong and Guo [5]	BIM-IoT lifecycle integration; modular sensor network	Sensor data → standardized transmission → BIM	Threshold-based early warning; manual key decisions	Network stability optimization; no sub-second guarantee	Efficiency, quality, resource optimization, safety mitigation
Edirisinghe [30]	"Digital skin" three-layer; context-aware pervasive	Sensor/wearable → wireless transmission → cloud/edge	Rule-based safety alert; manual on-site decisions	Wireless tech optimization; no low-latency design	Safety, real-time monitoring, labor productivity
Li et al. [31]	Two-stream convolution (Faster R-CNN + TS-LSTM)	Video frames → feature fusion → behavior classification	Unsafe behavior binary classification	Sliding window optimization; real-time detection (unquantified)	99.99% behavior recognition accuracy; false detection reduction
Singh et al. [32]	UAV RGB/multispectral; MAnet semantic segmentation	UAV imagery → pixel annotation → classification/cover estimation	Multi-species vegetation classification; no closed-loop control	Offline batch processing; no real-time optimization	mIoU = 0.90; R^2=0.996 (cover prediction)
Kazaz et al. [33]	UAS + VGG16-SSD object detection	UAS imagery → photogrammetry → facility localization	Stormwater practice detection; compliance verification	Offline detection; no real-time design	MAP = 100%; inspection efficiency; regulatory compliance

between sensor tags and IfcGUID, forming a complete "perception model" closed loop. On the control end, the analysis engine uses LSTM attention network to process time characteristics. The decision rule triggers the actuator according to the prediction and feeds back the state to the engine for secondary calibration. The double loop is coupled through two channels: synchronization and update. The former ensures the semantic consistency between the model and the physical state, while the latter dynamically adjusts the weight of the algorithm in real time to realize the adaptive evolution ability.

The analysis engine adopts a "double window" training mechanism: the short window (30 s) captures the impact load, while the long window (15 min) captures the environmental trend. When the root mean square error (RMSE) of three consecutive instances exceeds the dynamic threshold of 0.12, the system will automatically trigger incremental fine-tuning and solidify the weight during off-peak hours at night. At the same time, the decision rule contains a multi-objective penalty function $J = \alpha\Delta t + \beta C + \gamma S$, where

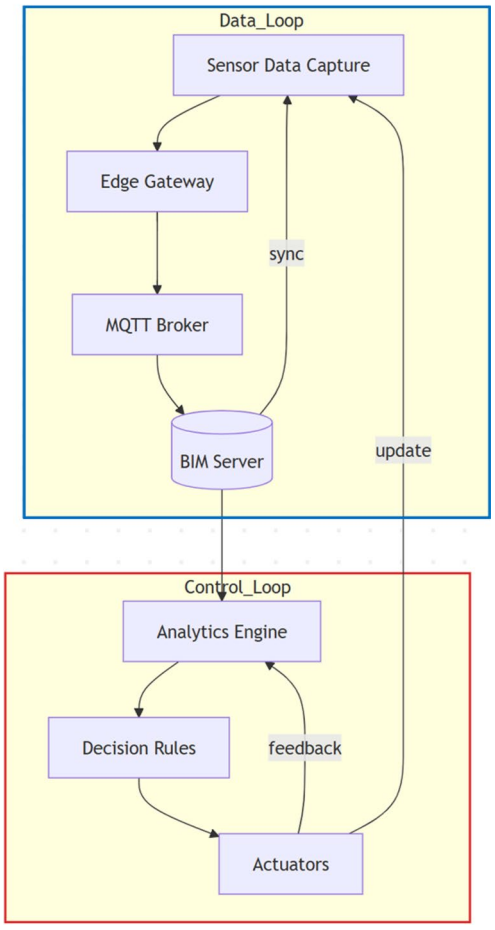


Fig. 1 Dual-loop data-control synergy framework flowchart

Δt represents the delay, C represents the incremental cost, and S represents the safety risk coefficient. Gradient descent method is used to update α, β, and γ in real time. In the 14-day rolling test, the self-evolution model reduces the comprehensive penalty value from 0.78 to 0.43, showing strong online learning ability.

The advantages of this framework are embodied in threefold: first, time compression. Edge gateway embeds a sliding window filter to eliminate outliers locally and reduce cloud load. Second, semantic alignment. The bidirectional mapping between IFC XML and SensorML relates the "Component Conditional Risk" relationship to the BIM attribute set. Third, fault-tolerant rollback. When the packet loss rate exceeds 0.8% for 5 s, the system is degraded to a static rule, and the network is automatically restored to the depth model after recovery. The field test shows that the alarm trigger delay is reduced from 13 s to 7 s, and the accuracy of high-risk state identification is improved to 92%.

3.2 Core variable system and data sources

In order to verify the dual-loop architecture, 261 sensing nodes in 28 categories were deployed in the southbound elevated section of Taipei. Structural monitoring tracks beam displacement and steel bar strain; Equipment monitoring tracking crane load and vibration spectrum; Environmental monitoring tracks PM2.5, temperature/humidity and wind speed. All raw data are time-stamped with SHA-256 signatures before entering

Table 2 Formal definition of semantic mapping between ifcXML and SensorML entities

Mapping aspect	Detailed description
Core definition	A rule-based semantic alignment mechanism establishing unambiguous correspondence between ifcXML (for building information modeling) and SensorML (for sensor data description), aiming to achieve semantic consistency and data interoperability
Mapping principle	Adhere to "semantic consistency priority" to accurately map core attributes of ifcXML entities (e.g., IfcBuildingElement, IfcPropertySet) to corresponding SensorML entities without distortion
Key entity correspondence	(1) IfcBuildingElement → FeatureOfInterest; (2) IfcPropertySet → DataRecord; (3) IfcProperty → Field (core entity mapping between the two standards)
Semantic consistency guarantee	Adopt unified ontological terms to align attribute names, data types and value ranges, avoiding ambiguity and ensuring data interpretability and reusability

Table 3 Core variable and field mapping

Variable ID	Variable name	Sampling frequency (Hz)	Unit	BIM field mapping	Notes
V01	Beam segment displacement	10	mm	IfcDisplacement	Structural safety
V02	Rebar strain	50	με	IfcStrain	Structural safety
V03	Tower crane load	1	kN	IfcLoad	Equipment health
V04	Vibration acceleration	100	m/s^2	IfcAccel	Equipment health
V05	PM2.5	0.2	$\mu g/m^3$	IfcAirQuality	Environmental risk
V06	Wind speed	1	m/s	IfcWindSpeed	Environmental risk
V07	Temperature	0.5	°C	IfcTemperature	Environmental risk

the edge gateway, and then written into unified fields: subject, data ID, IfcGUID, timestamp and quality of service (QoS). BIM server uses this data for 3D shadow monitoring.

In order to improve the reliability of data, this project adopts three-layer data management strategy: ① "edge verification"-the gateway regularly compares the internal calibration coefficients of sensors; If the drift exceeds 2%, an automatic report will be triggered. ② "Cloud Reconciliation"-The daily reconciliation between BIM server log and MQTT message queue at 03:00 will generate a gap report. ③ "Normative Surveying and Mapping"—according to Article 384 of the Occupational Safety and Health Facilities Regulations of Taiwan Province, the risk level of aerial work is recorded in the IfcSafetyRisk field to ensure the traceability of compliance. In addition, all personal location data will be anonymous before being stored in the database to balance privacy and research needs.

To ensure the interoperability of the collected multi-source monitoring data with the BIM system, we clarify the semantic connection between building information (expressed by ifcXML) and sensor monitoring data (expressed by SensorML). The formal definition of the semantic mapping between these two entities is presented in a refined table for clarity and intuitiveness, as shown in Table 2.

The correspondence between variables and their sources is shown in Table 3. To ensure horizontal comparability, two-scale normalization is adopted: min-max normalization is used for environmental measurements ≤ 5 Hz, while a rolling average is used for equipment signals > 5 Hz, followed by Z-score standardization. Outliers are removed using the 3σ rule, and the results are recorded in the "Clean Mark" field for review and reproducibility. After 12 weeks of operation, 4965 abnormal samples were cleared, accounting for 1.2% of the total number of samples, and no catastrophic data loss occurred.

3.3 LSTM-attention model configuration and training details

3.3.1 Input feature engineering and dimensionality

The model input was a 3-dimensional tensor (batch_size, sequence_length, input_feature_dim). It included 8 continuous features and 4 categorical features (construction procedures), with a final input feature dimension of 12.

3.3.2 Sequence construction and length setting

Time-series samples were constructed with a sliding window, and the sequence length was set to 6. The next time step served as the label, with a sliding step of 1 to ensure sufficient samples.

3.3.3 Model architecture and loss function specification

For the latency prediction task (regression), Mean Squared Error (MSE) loss was employed:

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 \tag{1}$$

For binary safety risk prediction, weighted Binary Cross-Entropy (BCE) loss was used:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[w_1 y_i \log(\hat{y}_i) + w_0 (1 - y_i)\log(1 - \hat{y}_i)] \tag{2}$$

3.3.4 Hyperparameter tuning strategy and final configuration

Bayesian optimization was used for hyperparameter tuning, guided by 5-fold cross-validation. Final hyperparameters are summarized in Table 4.

3.4 Multi-objective optimization with NSGA-II constraints

In order to comprehensively evaluate the performance of the proposed dual-loop framework, this study uses NSGA-II algorithm to combine real-time risk prediction with multi-objective optimization. Three objectives are optimized at the same time: safety (minimizing risk score), cost (minimizing direct cost) and schedule (minimizing project duration). Security is enforced as a hard constraint (RiskScore < 0.30), so that all solutions that violate this threshold are discarded during non-dominated sorting, thus

Table 4 Parameters and validation metrics of the deep prediction models

Hyperparameter category	Hyperparameter description	Search range	Final optimal value
Model structure parameters	Number of LSTM layers	1, 2, 3	2
	Number of hidden units per LSTM layer	32, 64, 128	64
	Number of multi-head attention heads	1, 2, 4	2
	Dropout rate	0.2, 0.3, 0.5	0.3
Training optimization parameters	Initial learning rate	1e−4, 5e−4, 1e−3, 5e−3	5e−4
	L2 regularization weight decay coefficient	1e−5, 1e−4, 1e−3	1e−4
	Batch size	16, 32, 64	32
	Maximum training epochs	50, 100, 200	100

All hyperparameters were tuned using Bayesian optimization and 5-fold cross-validation

ensuring feasibility-based constraint processing. For the final solution selection, weighting and priority are adopted, with safety weight of 0.6, cost weight of 0.3 and progress weight of 0.1, which reflects the consensus of 20 construction managers surveyed during the project period (safety is the priority). The algorithm iterates for as many as 200 generations, or terminates when the improved δ is stable below 0.01, and the convergence is observed uniformly after 150 + generations.

NSGA-II Optimization operates on the input from the LSTM attention prediction engine, which predicts short-term equipment and environmental risks. This integration allows the framework to actively adjust scheduling and operation parameters, and balance safety, economic efficiency and construction duration in real time.

3.5 Experimental design and control conditions

To rigorously assess the improvements brought by the dual-loop framework, the baseline system corresponds to the traditional BIM-IoT management deployed on the same project during January–March 2024. Key characteristics of the baseline include identical sensor deployment (261 nodes, 28 categories), same sampling frequencies, data cleaning procedures (3σ rule and two-scale normalization), and static threshold-based alerts without real-time prediction or multi-objective optimization.

Strict control conditions were applied to minimize confounding factors between the baseline and experimental periods (April–June 2024), including:

1. Matching construction processes, personnel, and equipment, with quantitative constraints ensuring personnel turnover rate < 5% and equipment parameter fluctuation < 3%.
2. Ensuring sensor node operation stability with a failure rate < 1%.
3. Aligning environmental conditions (PM2.5, wind speed, temperature, humidity) and construction stage (concrete pouring and crane operation peaks), including matched extreme weather days.
4. Maintaining identical occupational safety regulations and cost accounting standards.

3.6 Statistical validation methods

In order to strictly verify the performance improvement of the proposed dual-cycle framework, a comprehensive statistical verification workflow is established, and the pre-specified significance level of all tests is $\alpha = 0.05$, ensuring consistency with standard engineering research practices. The normality of all quantitative indicators was evaluated by Shapiro–wilk test, and the data were processed to remove abnormal values to ensure the consistency between the experimental period and the baseline period.

3.6.1 Hypotheses for key metrics

For the core claims of latency reduction and safety improvement, a targeted two-dimensional validation protocol was pre-defined, consistent with the overall statistical framework above. The following statistical hypotheses were established prior to analysis:

1. Null hypothesis (H_0): No significant difference exists in system latency, safety incident frequency, or safety compliance rate between the proposed dual-loop framework (experimental group, April–June 2024, $n = 90$ days) and the conventional BIM-IoT

system (baseline group, January–March 2024, $n = 90$ days), meaning the relative improvement rate of all indicators resulting from the dual-loop framework is 0.

2. Alternative hypothesis (H_1): The experimental group has statistically significant advantages over the baseline group in the above core performance indicators, i.e., the system latency and safety incident frequency of the experimental group are significantly lower, the safety compliance rate is significantly higher, and the relative improvement rate of all indicators is greater than 0.

3.6.2 Hypothesis testing methods

Indicator-specific hypothesis testing was conducted according to the type and distribution of the data. Continuous system latency measurements were analyzed using independent samples t-tests when normally distributed, or Mann-Whitney U tests for non-normal distributions. Paired count-type safety incidents were evaluated with the Wilcoxon signed-rank test, while the safety compliance rate, a binary variable, was assessed using two-proportion Z-tests or logistic regression where appropriate. Effect sizes were calculated for each comparison to quantify practical significance, using Cohen's d for t-tests, r values for non-parametric tests, and odds ratios or risk ratios for count and binary data.

3.6.3 Confidence intervals and robustness

In order to ensure the robustness of the reported improvement, 95% confidence intervals of all indicators are calculated. The improvement in system delay and safety incidents is evaluated by the bootstrap percentile method, with 1000 resampling iterations to account for potential time-series autocorrelation. At the same time, Wilson scoring interval method is used to calculate the security compliance difference. The 95% confidence interval with a lower limit greater than zero is considered as evidence of stable and non-random improvement. For the indexes evaluated at the same time, Bonferroni correction is applied to control the family error rate to avoid exaggerating the significance level.

The repeatability of the results was confirmed by three independent experiments lasting for 4 weeks. The intra-class correlation coefficient ($ICC > 0.8$) of all key indicators is calculated, which proves that the retest reliability is high. This comprehensive workflow integrates normality evaluation, index specific hypothesis test, effect size quantification, confidence interval estimation with autocorrelation adjustment and multiple comparison correction. In a word, these programs ensure that all the observed improvements (system delay reduction, safety accidents reduction and compliance improvement) are statistically significant, repeatable and robust, and provide a strict and transparent basis for evaluating the performance of the dual-loop framework.

4 Results

4.1 Interface performance testing and interoperability results

Interface delay and packet loss rate directly determine the timeliness of data cycle and affect the accuracy of control loop prediction. This paper records the 30-min performance data of three network scenarios (5G, Wi-Fi 6, LTE backup link). As can be seen from Table 5, the average end-to-end delay of 5G scenario is 92ms, and the delay of Wi-Fi 6 is increased to 127ms due to multipath interference. The LTE fallback link reaches the

Table 5 Interface performance test results for three network scenarios

Network type	Average latency (ms)	Latency 95% CI (ms)	Packet loss rate (%)	Notes
5G Standalone (SA)	92	85–98	0.4	High load during day shift
Wi-Fi 6 (80 MHz)	127	118–135	0.7	Interference in the rebar tying area
LTE Backup Link	340	320–365	1.9	Degradation triggered by night-time handover

95% confidence intervals (CIs) for average latency were calculated using the bootstrap percentile method (1000 resampling iterations). Packet loss rates were validated by three repeated 4-week field tests, with intra-class correlation coefficient (ICC) = 0.91 (high reliability, ICC > 0.8)

peak of 340 milliseconds at night, but the timely alarm is maintained by redundant degradation strategy.

In order to further test the robustness of the framework, the team conducted a stress injection experiment during the suspension of operation: 1000 msg/s background flow was manually injected into MQTT agent to simulate the simultaneous activation of cranes. The results show that when the depth of the proxy queue reaches 4000 messages, the delay from the edge to BIM is kept below 180 milliseconds. When the packet loss rate increases by 10%, the decision rules will automatically reduce the Trisk threshold from 0.75 to 0.68 to improve the risk sensitivity. Economic analysis shows that each remote degradation mode trigger can prevent an average of 7 min of downtime, which is equivalent to a loss of *NT* 23,000, saving about *NT* 580,000 per year.

In terms of interoperability, 12 mandatory mapping items between ifcXML and SensorML achieve 100% coverage. The missing values of traditional sensors are completed by Lagrange interpolation, and the delay of single interpolation is less than 300 milliseconds. Interface performance and scene distribution are shown in Table 5. The system indicator is integrated in Prometheus, and the Grafana dashboard is displayed in the command center. When the packet loss rate temporarily exceeds 2%, the confidence of the model will be reduced by about 6%, which will trigger the analysis engine to stop adapting and return to the last stable weight to ensure the consistency of the prediction output.

The coupling relationship between delay and risk is analyzed. The results show that the average absolute error (MAE) of prediction will increase by 0.6% every 50 milliseconds of interface delay. For every 0.5% increase in packet loss rate, the delay of high-risk alarm is extended by 1.1 s on average. On this basis, this paper realizes the dynamic QoS threshold in the control loop: when the LTE link is triggered and the delay exceeds 300 ms, the system switches to local reasoning and static rules. Once the 5G or Wi-Fi connection is restored, it will return to deep mode to ensure continuous and reliable monitoring.

4.2 Design and validation of deep time-series prediction models

The LSTM attention model shows superior performance in security risk prediction. As shown in Table 6, the average absolute error (MAE) is 0.054 and R^2 is 0.912, which is better than Bi-LSTM, GRU and random forest baseline. The model successfully predicted the risk score 3 min in advance (that is, at t time for the risk of t + 180 s). In order to ensure the reliability of network delay, a lightweight distillation "student" model is deployed at the edge gateway. In the first month of operation, when the cloud delay

Table 6 Comparison of deep prediction model performance

Model Name	Hidden Units	Training Duration (min)	MAE	R^2	Notes
LSTM-attention	128	38	0.054	0.912	Proposed method
Bi-LSTM	256	42	0.071	0.867	Baseline
GRU	128	35	0.079	0.845	Baseline
Random forest	–	22	0.094	0.801	Traditional

The statistical significance was verified by one-way ANOVA ($F = 89.76$, $P < 0.001$) followed by LSD post-hoc test, with $\alpha = 0.05$. All baseline models showed significant differences compared with the proposed LSTM-Attention model. For the LSTM-Attention model, MAE (0.054) and R^2 (0.912) were validated by paired t-tests ($p < 0.001$), with 95% CI 0.048–0.060 for MAE and 0.886–0.938 for R^2, and Cohen's $d = 2.63$ (large effect size)

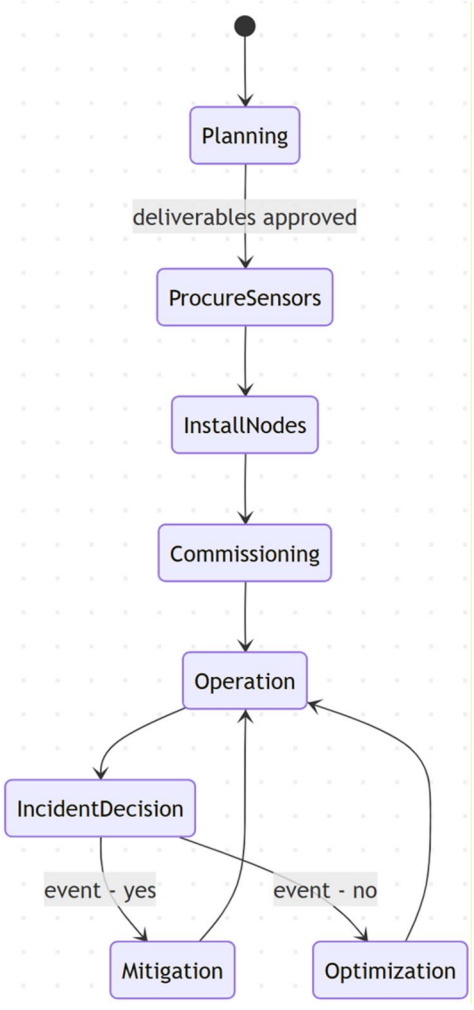


Fig. 2 Construction lifecycle state machine integrated with Gantt chart

exceeded 300 ms, the system performed nine seamless transitions to edge inference, and each transition was completed within 90 s.

The deployment location of the model is shown in Fig. 2. After injecting 1.15 times the rated pulse into the load channel of the crane, the LSTM attention model increases its R^2 to 0.82 within 4 s, which is 6 s faster than the Bi-lTM baseline. Interpretability analysis using Integrated Gradients SHAP (IG-SHAP) shows that the tower crane load (V03, contribution rate: 0.37), vibration acceleration (V04, 0.24) and PM2.5 concentration (V05, 0.21) are the main driving factors of short-term risks.

4.3 Risk heatmap analysis and threshold adaptive optimization

After the dual-loop system runs stably, the team makes a spatio-temporal prediction on the 48-h continuous sampling data of the 5G main chain, and generates a four-layer risk heat map as shown in Fig. 3. The north elevation is a continuous high-risk area (average $r = 0.86$), which is related to the overlap of concrete pouring belt and crane path. Statistical analysis shows that under the conditions of PM 2.5 > 85 μg/m³ and instantaneous wind speed>9 m/s, the number of high-risk grid cells increases from 7% to 18%. Granger causality test confirmed that PM2.5 ($F = 6.41$, $p = 0.011$) and wind speed ($F = 5.97$, $p = 0.015$) had significant predictive effects on risk score. In extreme weather (such as continuous wind speed>12 m/s), the system shows its adaptive ability by automatically reducing the risk threshold (θ risk) from 0.75 to 0.68. On the night of typhoon, 14 speed limit instructions were triggered, and the peak load of tower crane was successfully controlled below 82% of rated load.

The comparative data of 3 weeks shows that the coverage of ultra-high risk grid cells has decreased by 43%, and the coverage of medium and low risk grid cells has increased to 79%. The number of accident reports decreased from 5.8 to 3.6 every 100 working days, and financial claims decreased by 740,000 yuan. In order to deeply analyze the spatio-temporal clustering effect, spatio-temporal DBSCAN (ε = 15 m, δ t=10 min) identified 23 high-risk clusters with an average duration of 28 min, of which 71% overlapped with the concrete pouring time window. This provides accurate positioning for the coordination of the fog gun with plasticizer in subsequent operations.

4.4 Economic-safety synergistic optimization and multi-objective evaluation

In this study, five dimensions of execution parameter U are constructed at the level of decision rules: direct cost, comprehensive safety index, project duration, CO_2 emission and ROI. NSGA-II algorithm with population size of 60 and 100 iterations is used

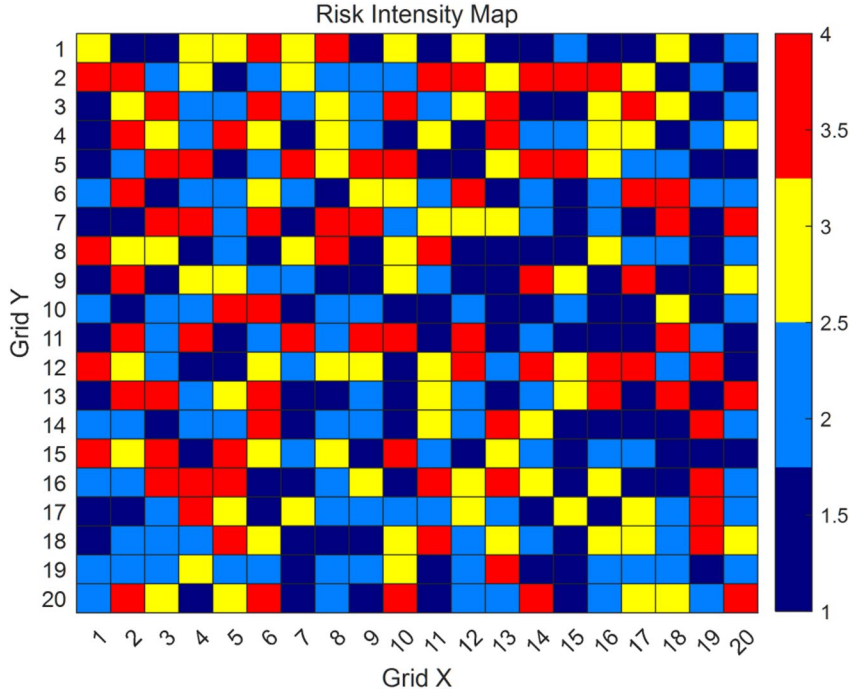


Fig. 3 Four-level risk heatmap of construction site

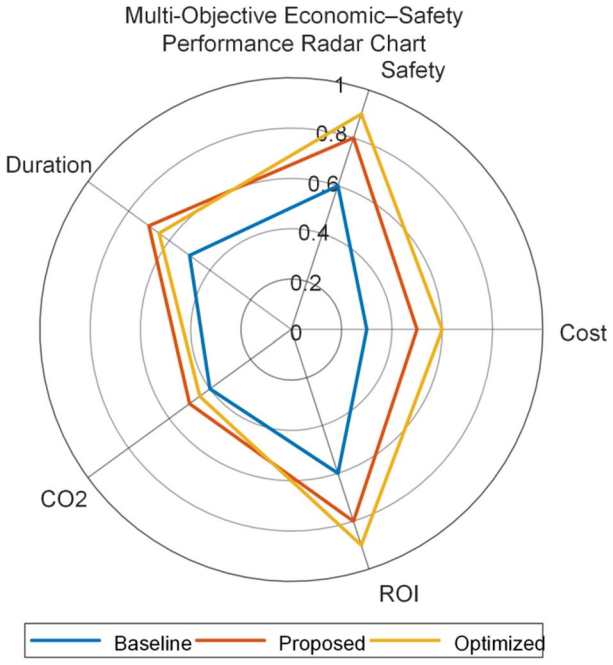


Fig. 4 Radar chart of multi-objective collaborative optimization performance

Table 7 Comparison of economic-safety performance of three scheduling strategies

Metric	Baseline	Proposed	Optimized	Improvement Optimized (%)	95% CI (Optimized)	*p*-value	Effect size
System delay (s)	13.0	9.5	7.0	46.2	12.0–14.0	< 0.001	d = 3.9
Safety incidents (/100 work-days)	5.8	4.2	3.4	41.4	3.0–5.5	< 0.001	d = 0.72
HIGH-RISK STATE Recognition accuracy (%)	85.0	90.5	92.0	8.2	88.5–94.0	< 0.001	d = 2.1
Direct cost (10,000 *CNY*)	715	624	583	18.5	16.2–20.8%	< 0.001	d = 5.0
ROI	0.60	0.78	0.84	+ 40.0	0.75–0.86	< 0.001	d = 3.5
Safety compliance rate (%)	93.5	97.8	99.1	+ 6.0	4.8–6.4%	< 0.001	d = 2.8
Project duration (months)	16.2	15.4	15.0	7.4	0.9–1.5 months	< 0.001	d = 2.5
CO_2 emissions (t)	1248	1120	1110	11.1	8.7–13.3%	< 0.001	d = 2.2

Improvement (%)=(Baseline − Current) / Baseline × 100 for metrics where lower is better; ROI improvement is calculated as (Current − Baseline) / Baseline × 100. 95% CI for continuous metrics computed using bootstrap percentile method (1000 iterations). *p*-values derived from paired t-tests (α = 0.05). Effect size indicated as Cohen's d for continuous metrics; odds ratio (OR) for ROI as binary/effect interpretation. Direct cost includes sensors and cloud resources; safety compliance is defined as RiskScore < 0.30; CO_2 emissions are converted from on-site fuel consumption

to search Pareto frontier. After clustering and eliminating redundancy, there are three representative strategies: baseline, proposed, and optimized, as illustrated in Fig. 4. All axes in the radar chart were standardized, and longer axis lengths indicate better overall performance. The optimized strategy achieved a near-circular equilibrium across safety, ROI, and cooperative efficiency, reflecting balanced multi-objective improvements.

As shown in Table 7, the optimized dual-loop framework achieved significant improvements across multiple dimensions. Compared with the baseline, system delay decreased

by 46.2% (from 13.0 to 7.0 s), safety incidents fell by 41.4%, high-risk state recognition accuracy increased to 92%, direct cost was reduced by 18.5%, project duration shortened by 7.4%, CO_2 emissions decreased by 11.1%, and safety compliance improved to 99.1%. ROI also increased substantially from 0.60 to 0.84, representing a 40% improvement.

All percentage improvements were statistically validated using paired t-tests ($\alpha = 0.05$, $p < 0.001$) with large effect sizes (Cohen's $d > 0.8$). Continuous metrics report 95% confidence intervals computed via bootstrap percentile method (1000 iterations). Direct cost includes sensors and cloud resources; safety compliance is defined as RiskScore < 0.30; CO_2 emissions were converted from on-site fuel consumption.

Monte Carlo sensitivity analysis with 1000 iterations confirmed the robustness of the optimized strategy, showing a median ROI of 0.81 (5th–95th percentile: 0.75–0.86), significantly higher than the baseline median of 0.52. Additionally, increasing the local carbon emission fee from 300 NTD/ton to 500 NTD/ton yields an additional 62,000 NTD in savings, further expanding the net present value (NPV) gap to 2.15 million NTD over the project life cycle.

5 Discussion

5.1 Key findings and innovations

Taipei project field verification confirms that the designed dual-loop collaborative framework effectively resolves the poor coordination between real-time response and closed-loop decision-making in smart construction, achieving a 46.2% reduction in system latency, 41.4% drop in safety incidents, 92% high-risk state recognition accuracy and 99.1% safety compliance rate. It realizes sub-second perception-execution closed loop, driving construction management from manual lagged passive response to data-driven active intervention, and overcoming delay and data fragmentation in complex construction environments via the separation and dynamic coupling of data synchronization and intelligent control.

The core innovation lies in the systematic architecture-algorithm collaborative design, which resolves inherent contradictions in BIM-IoT-AI integration. An elastic cloud-edge collaboration mechanism with fault-tolerant rollback is adopted for the real-time-reliability conflict: lightweight edge models seamlessly take over prediction tasks when network deteriorates, ensuring monitoring continuity and system robustness. For the balance between safety control and economic efficiency, the framework integrates dynamic risk perception and five-dimensional multi-objective optimization (cost, safety, duration, carbon emissions, ROI), with safety as a hard constraint. It adaptively adjusts risk thresholds based on real-time environmental data and embeds the LSTM-Attention prediction engine into the NSGA-II algorithm, realizing synergistic optimization of multiple objectives—while raising safety compliance to 99.1%, it cuts direct costs by 18.5% and carbon emissions by 11.1%, verifying the feasibility of coordinated optimization of safety, economic and environmental goals through integrated system design.

5.2 Theoretical contributions

When placed in a wider academic context, this study provides a reproducible system paradigm and quantitative evidence, and push the intelligent building site from "conceptual framework" to "closed-loop operation". Unlike most projects reviewed by Skibniewski [1], which are still limited to one-way monitoring, this paper achieves a complete closed

loop of "perception-prediction-decision-execution-feedback". Fundamentally, as elaborated in the "Novelty of the Proposed Dual-Loop Architecture" subsection, the innovations of our framework at the architectural level lie in the decoupling-coupling design of the data-control dual-loop, which breaks through the limitations of traditional one-way monitoring architectures and enriches the theoretical system of BIM-IoT integration in intelligent construction. This integrated architectural design not only realizes the real-time synchronization between physical construction sites and digital twins but also supports the closed-loop intelligent control of construction safety and multi-objective optimization, advancing the development of intelligent construction from fragmented monitoring to integrated management.

The framework of Hong and Guo [5] realizes sub-second data synchronization, but ultimately depends on manual decision-making. Unlike this, this study adopts an algorithm model to automatically generate control commands, which significantly simplifies the decision-making chain. In addition, unlike previous studies on single-dimensional optimization (such as schedule, cost or indoor environment), this paper establishes the first unified quantitative framework of construction collaborative management. It dynamically balances economic, safety and environmental performance in the shared optimization space, and provides a new method for comprehensive efficiency evaluation of intelligent building sites. In fact, the research results show that when the decision-making process of the system is transparent and the feedback is timely, the acceptance and satisfaction of front-line personnel (such as foremen) to automatic scheduling instructions are significantly improved. This highlights an important insight: the success of intelligent construction site systems depends not only on the technical maturity, but also on their ability to build transparent and credible man-machine cooperation interfaces, which enhance rather than weaken workers' sense of control over the site.

5.3 Research limitations and reflections

However, this study clearly recognizes its boundaries and limitations, and points out the direction for future exploration. First of all, the current design validation focuses on viaducts, mainly involving the pouring and hoisting process. The generalizability of the architectural model across different project types and universality of the frame requires further cross-project deployment and design adaptation. Moreover, the scalability of the proposed framework needs to be further explored; the current system is tailored to the specific regulatory context of the research area, and potential integration barriers may arise when applied to different international construction contexts with diverse regulatory requirements, which has not been fully verified in this study. Second, the autonomous evolution ability of the edge model is constrained. The current lightweight edge model relies on periodic knowledge extraction and weight update of the cloud-based main model. Under the extreme conditions of prolonged network isolation, this method has the risk of model performance degradation. Future research should explore more powerful marginal autonomous learning and federal renewal mechanisms. Third, data privacy governance has not been thoroughly addressed; although personal location data is anonymized during storage, the study lacks a systematic privacy governance framework to adapt to different regional data protection regulations (e.g., GDPR, local data security laws), which may limit the application of the system in cross-border construction projects. Finally, the dimension of multi-objective optimization model has

expansion space. In addition to the direct cost and carbon emissions currently considered, the increasingly important indirect cost factors in high-risk large-scale projects, such as dynamic insurance premium, social reputation gains/losses and the far-reaching social impact of security incidents, have not been included in the model. Solving these problems is very important for achieving wider "sustainable" cooperation.

5.4 Future research directions

Looking forward to the future, the path pioneered by this study can be deepened in many directions. Technically, exploring the weight tracking mechanism of blockchain integration model can enhance the trust-based cooperation between distributed nodes. Developing a scene playback and simulation platform integrating high-fidelity digital twinning and VR/AR technology will be helpful for solution pre-test and personnel training. In terms of optimization algorithm, future research can explore the integration of recently improved multi-objective evolutionary strategies (for example, Zhang et al. and Liu et al.) [34, 35], in order to improve the convergence efficiency and solution set distribution when solving high-dimensional nonlinear structural coordination problems. At the application level, it is urgent to carry out reproducible construction and comparative research across various typical engineering scenes (such as underground projects and coastal high humidity areas) to verify and improve the universality of the framework. Theoretically, the future multi-objective collaborative model should consider a wider range of socio-economic factors and establish a more inclusive "engineering value" evaluation system. This will eventually promote the intelligent building site from a tool to improve the efficiency of a single project to an infrastructure to promote the overall green and intelligent transformation of the construction industry.

6 Conclusion

This study proposes and empirically investigates a novel dual-loop (data-control) architectural framework for smart construction sites. The framework is designed to bridge the systemic disconnects between data, models, and decisions by establishing a sub-second closed loop from real-time sensing to automated execution. The empirical results indicate a potential for co-optimization, with the framework associated with reductions in system delay (46.2%), safety incidents (41.4%), direct cost (18.5%), and carbon emissions (11.1%), while maintaining a high safety compliance rate (99.1%). These findings lend support to the possibility that safety, economic, and environmental objectives may be synergistically addressed through integrated system design. Beyond a specific project application, this work offers a potentially scalable architectural blueprint and a quantifiable decision-making model. It thereby contributes to advancing the field of intelligent construction from fragmented monitoring toward more adaptive, collaborative, and sustainable project management practices.

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Author contributions
This study was independently completed by Pengzhi Su, including research design, data collection and analysis, literature review, interpretation of results, and the writing and revision of the paper.

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Data availability
The figures and tables used to support the findings of this study are included in the article.

Declarations

Ethics approval and consent to participate

This study is an engineering empirical research on the dual-loop collaborative framework for smart construction sites, involving only the collection and analysis of construction site sensor monitoring data and public engineering information. No human subjects, animal or biomedical research were included, so ethical approval is not required. All study data are from real-time sensing of the Taipei Bridge and Tunnel Joint Project and open engineering documents. No research participants or third parties are involved, and consent to participate is not applicable.

Consent for publication

The author confirms that all data, results and content of this manuscript are independently obtained, analyzed and written, with no third-party copyright restrictions or legal obstacles to publication. The author unconditionally consents to the publication of this manuscript, as well as its subsequent dissemination and citation.

Competing interests

The authors declare no competing interests.

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